

Traffic Sign Recognition

Using Semi-Supervised Learning with Virtual Adversarial Training

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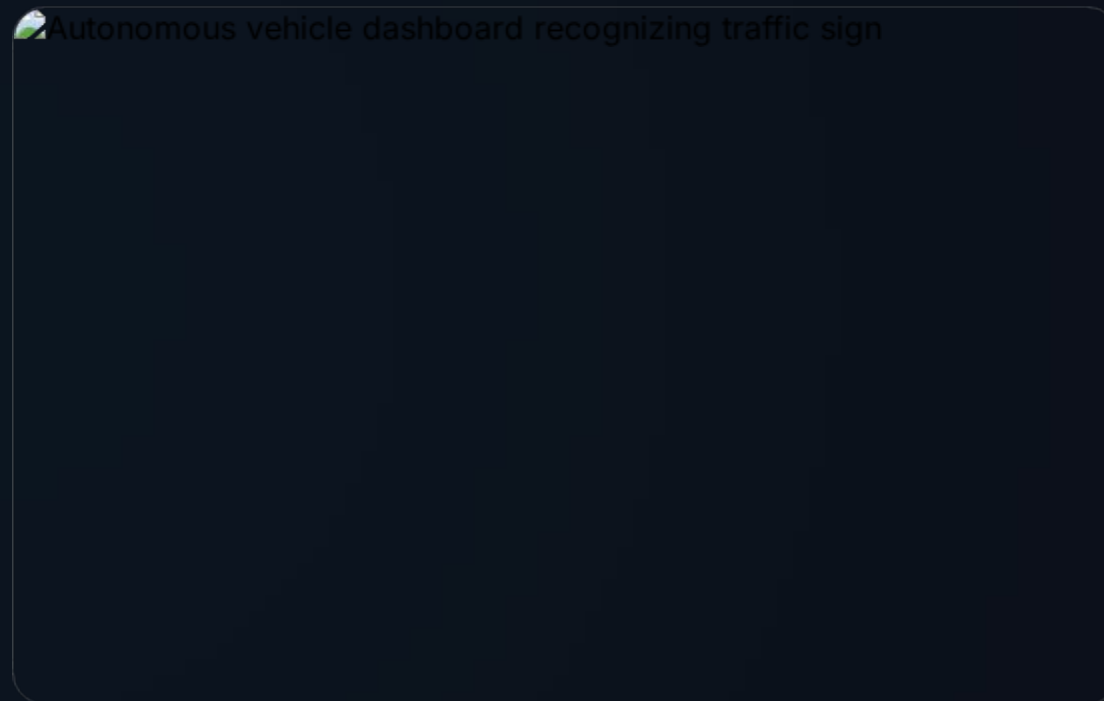
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Introduction & Context

The role of Machine Learning in modern autonomous driving systems.

The Rise of Autonomous Driving

- Traffic sign recognition is a fundamental, safety-critical component of modern autonomous driving systems.
- Vehicles must accurately identify and interpret signs in real-time to make safe decisions.
- Failure to recognize signs accurately can result in severe consequences, such as ignoring speed limits or failing to yield.
- Traditional Supervised Learning (SL) offers strong performance but relies heavily on vast amounts of strictly labeled data.



Machine Learning Paradigms



Supervised Learning

Models train exclusively on fully labelled data. The objective is to learn a direct mapping to predict outputs for new, unseen data accurately.



Unsupervised Learning

Works strictly with unlabelled data. The algorithm independently discovers hidden structures, patterns, or relationships (e.g., clustering).



Semi-Supervised

A powerful hybrid approach combining a small subset of costly labelled data with a vast, easily obtainable pool of unlabelled data.

The Annotation Bottleneck

In real-world autonomous driving, collecting and manually annotating massive datasets is notoriously expensive and time-consuming.

Semi-Supervised Learning (SSL) offers a critical solution, leveraging a small amount of labeled data alongside a vast pool of raw, unlabelled samples to drastically reduce human annotation effort.



Theoretical Foundations

Understanding Semi-Supervised Learning and Virtual
Adversarial Training.

Formulating Semi-Supervised Learning

The Labelled Set

In SSL, we begin with a traditional supervised dataset, containing input samples paired with their correct target labels.

$$D_L = \{ (x_i, y_i) \}_{i=1}^L$$

The Unlabelled Set

We introduce an additional, much larger collection of samples without labels. SSL aims to extract structural info from this distribution.

$$D_u = \{ x_i \}_{i=L+1}^{L+U}$$

Core SSL Assumptions



Smoothness

Data points that reside close to each other in the input space have a high probability of sharing the exact same label.



Low-Density

An optimal decision boundary should pass through low-density regions, avoiding cutting through densely populated clusters of samples.



Manifold & Cluster

High-dimensional data naturally lies on lower-dimensional manifolds, and grouped clusters are expected to belong to a unified class.

Virtual Adversarial Training



VAT improves local distributional smoothness
by training the model to resist the most
harmful small perturbation around each data
point, without requiring label information.



— Based on Miyato et al. (2015)

How VAT Works

VAT aims to make a model's output stable against small input perturbations, ensuring Local Distributional Smoothness (LDS).

It calculates the virtual adversarial perturbation that maximizes the Kullback-Leibler (KL) divergence between the original prediction and the perturbed prediction:

$$r_{\text{adv}} = \underset{r}{\operatorname{argmax}} \operatorname{KL} [p(y|x, \theta) \parallel p(y|x+r, \theta)]$$

The model is trained to minimize this divergence, smoothing the decision boundary effectively.



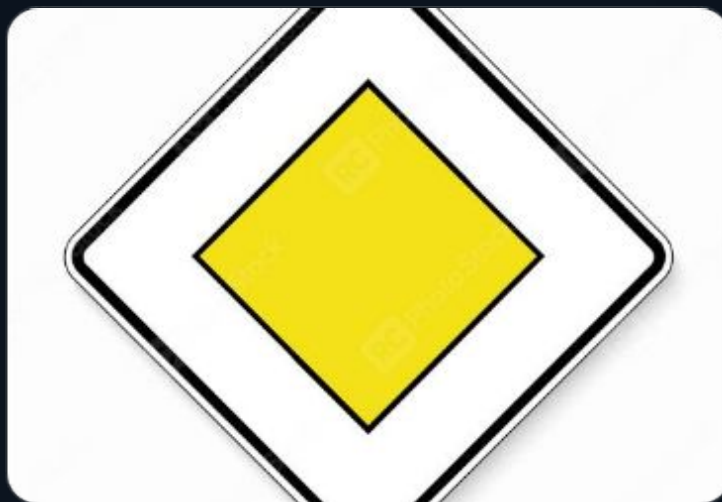
Methodology & Architecture

Datasets, Preprocessing, and Neural Network Models.

The **GTSRB** Dataset



Scale & Diversity: Over 50,000 traffic sign images spanning across more than 40 distinct traffic sign classes.



Real-world Variables: Includes heavy variations in illumination, angles, scale, rotation, and complex backgrounds.



Benchmark Status: A widely recognized gold standard for testing deep learning models in autonomous driving tasks.

Data Preprocessing Pipeline

-  **Standardization:** Every raw image is meticulously resized to a consistent 48×48 pixel dimension, a strict requirement for neural network processing.
-  **Geometric Augmentation:** During training, images are randomly rotated up to 15 degrees, shifted, zoomed or flipped to simulate imperfect real-world viewing angles.
-  **Color Augmentation:** Brightness, contrast, saturation and hue are dynamically adjusted, successfully mimicking night driving, shadows or harsh sunlight.
-  **Normalization:** Pixel values are mathematically adjusted to fit within a standard scale utilizing the specific average colors of the GTSRB dataset, speeding up computation.

Network Architectures

Fully Connected Network (FCN)

A traditional baseline architecture. It processes the $48 \times 48 \times 3$ image by completely flattening it into a single one-dimensional array of 6,912 features.

Utilizes two substantial hidden layers (1,200 neurons each) equipped with Batch Normalization, ReLU activation and Dropout before final output.

Small Convolutional Neural Network (SCNN)

Purpose-built for spatial image processing. It deploys three dedicated Convolutional Blocks acting as visual filters to detect edges and complex shapes. Each block pairs two convolutional layers with Max Pooling to extract critical features efficiently, culminating in a linear classifier.

The Training Pipeline

2. Training Loop

Model processes mixed batches of labeled and unlabeled data, updating parameters via VAT and supervised loss.

4. Early Stopping

If validation accuracy stalls for 15 consecutive epochs, training halts to preserve the best checkpoint state.

1. Configuration

Optimizer (AdamW) with weight decay, paired with Cosine Annealing scheduler and Cross-Entropy loss.

3. Validation

Performance evaluated on clean, unaugmented validation images to track true generalization and avoid overfitting.

Experiments & Results

Evaluating VAT impact across varying supervision levels.

Dataset Partitioning Strategy

GSTRB:

Train Directory (39,209 total images) split into:

85% Training (33,328 samples) - 64.3%

15% Validation (5,881 samples)- 11.3%

An 85/15 split maintains a robust pool for semi-supervised training while reserving a strict, unaugmented validation block to prevent overfitting and ensure objective accuracy tracking.

Test Set is loaded separately from the official GTSRB test files (12,630 samples) - 24.4% of the combined dataset.

Artificial:

1000 Artificial Samples

70% for Training

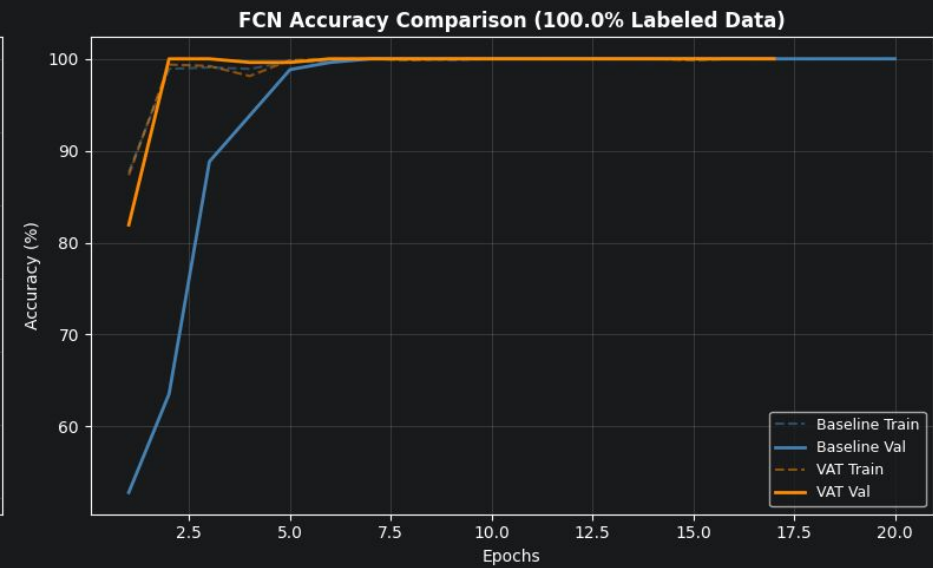
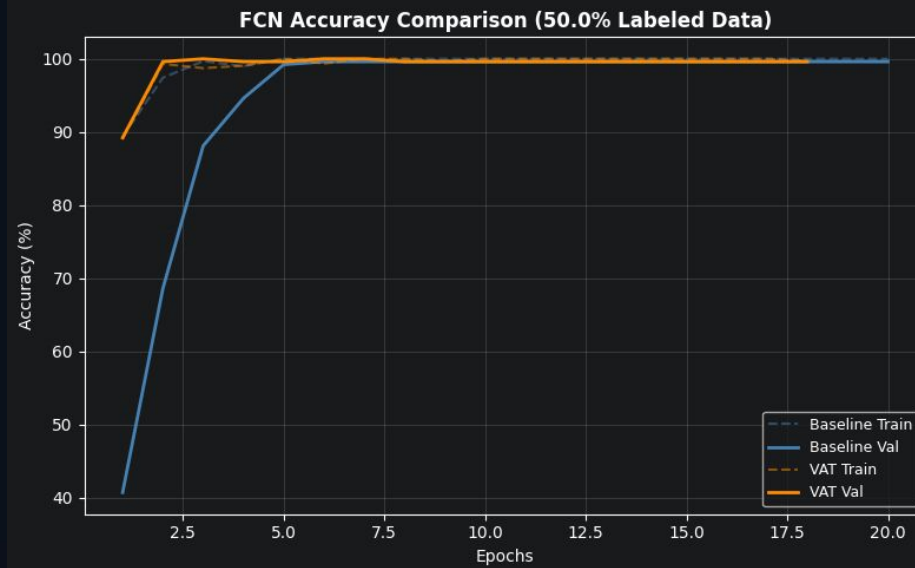
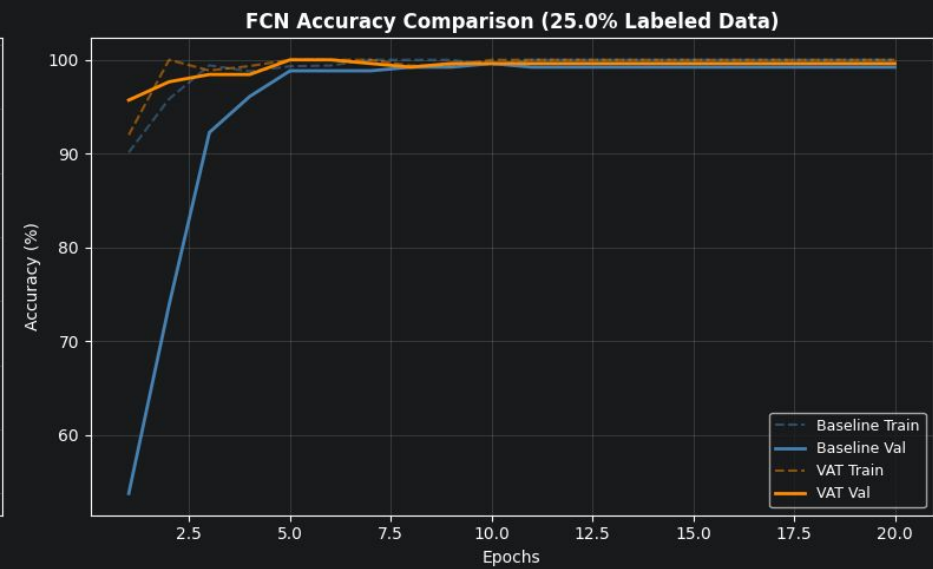
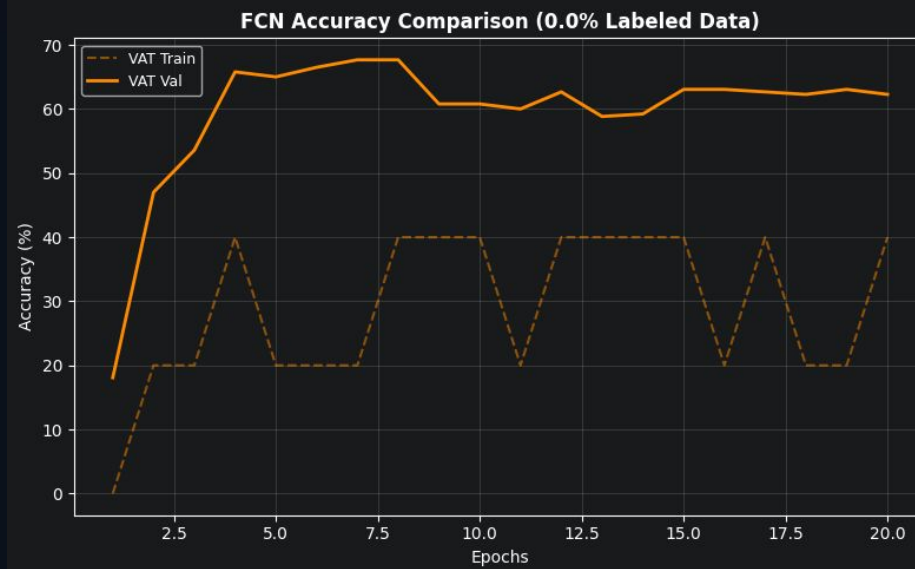
15% for Validation

15% for Testing



Artificial Environment: 0%, 25%, 50% and 100% Labeled Data

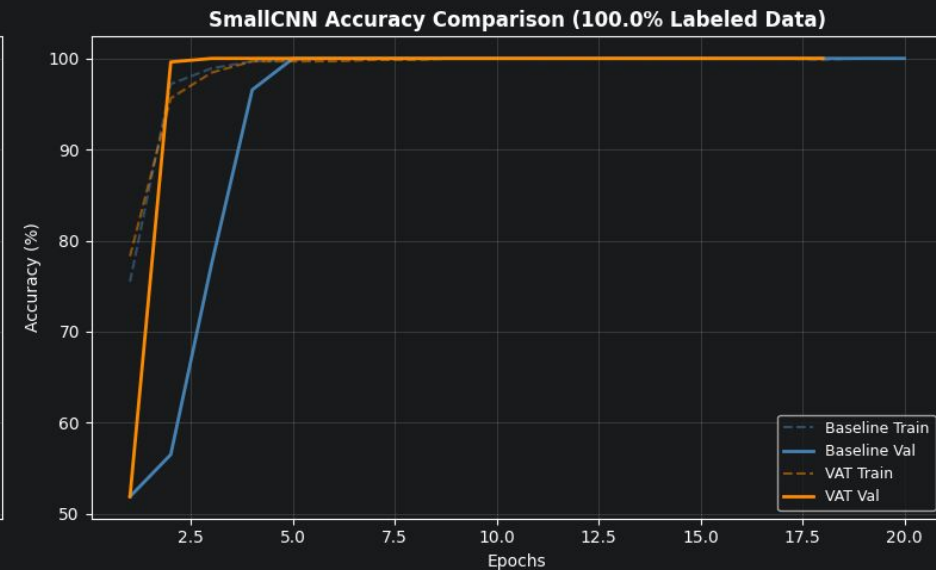
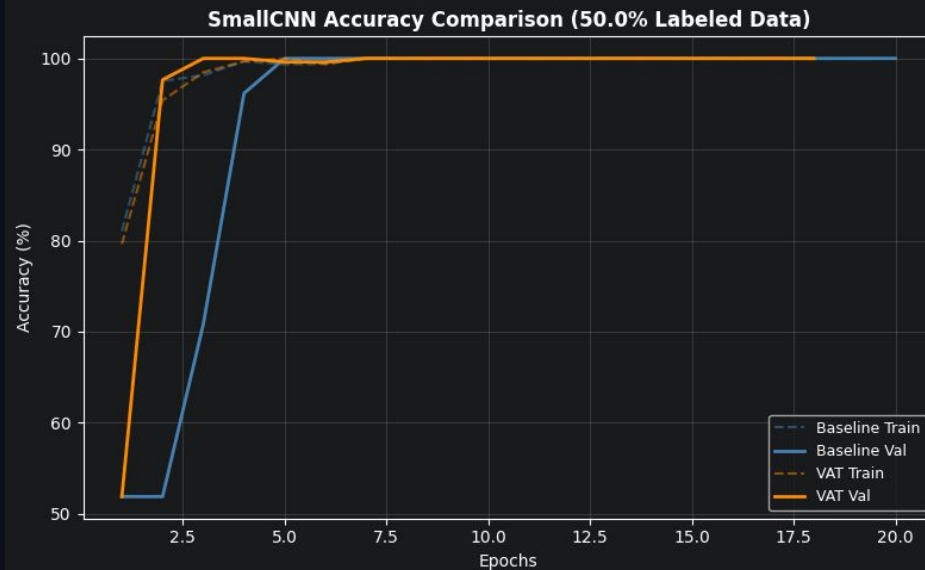
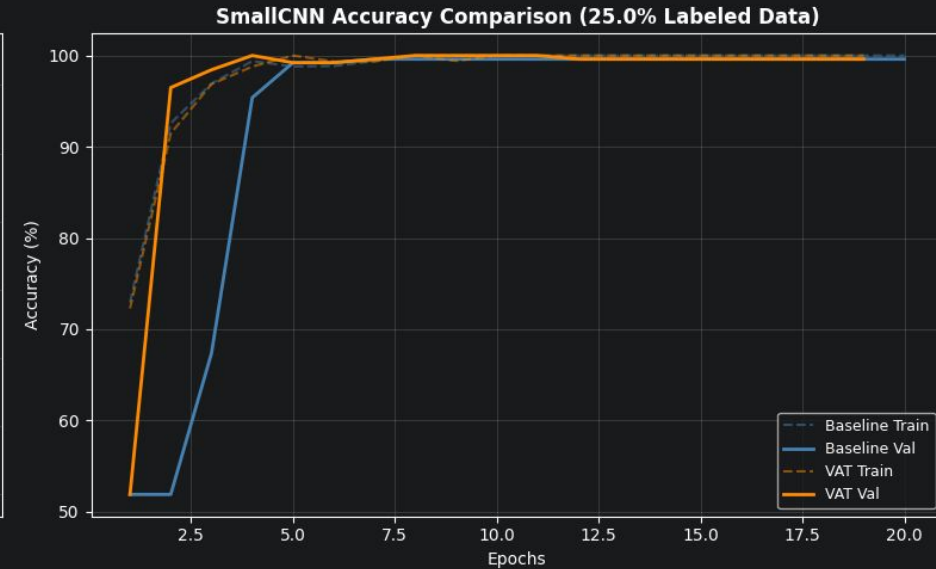
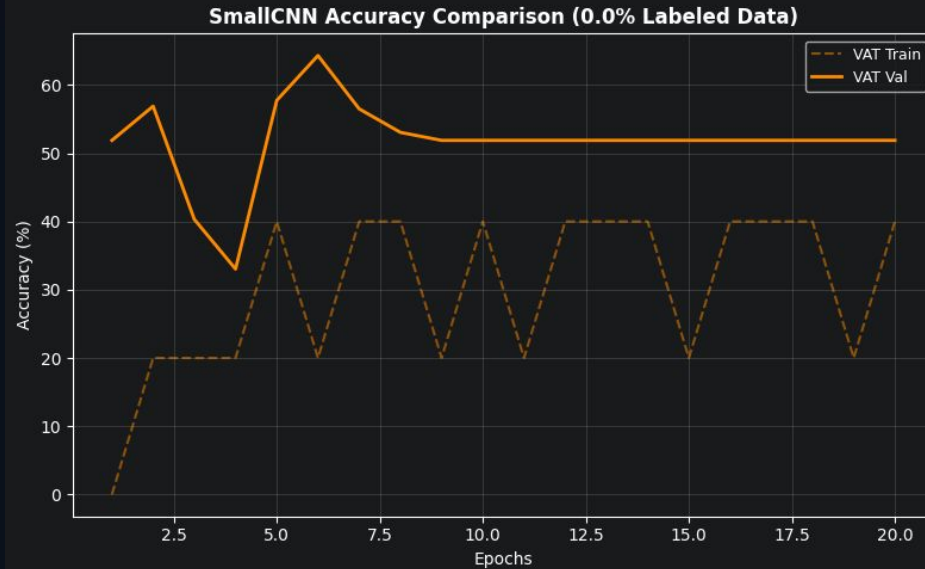
FCN



In an artificially controlled environment completely devoid of labels, only the VAT models achieved meaningful decision boundaries, proving VAT's ability to learn strictly from data distribution structures.

Artificial Environment: 0%, 25%, 50% and 100% Labeled Data

Small
CNN



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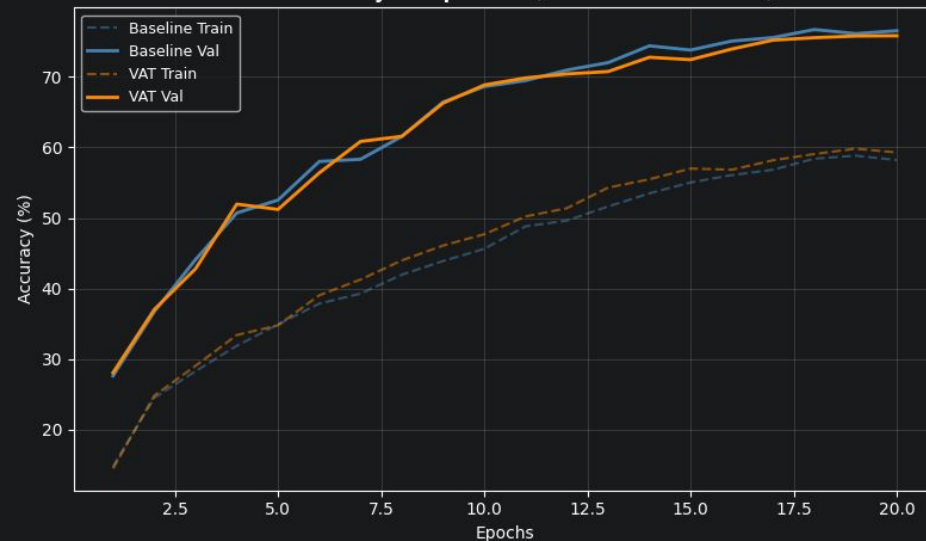
GTSRB: 0%, 25%, 50% and 100% Labeled Data

FCN

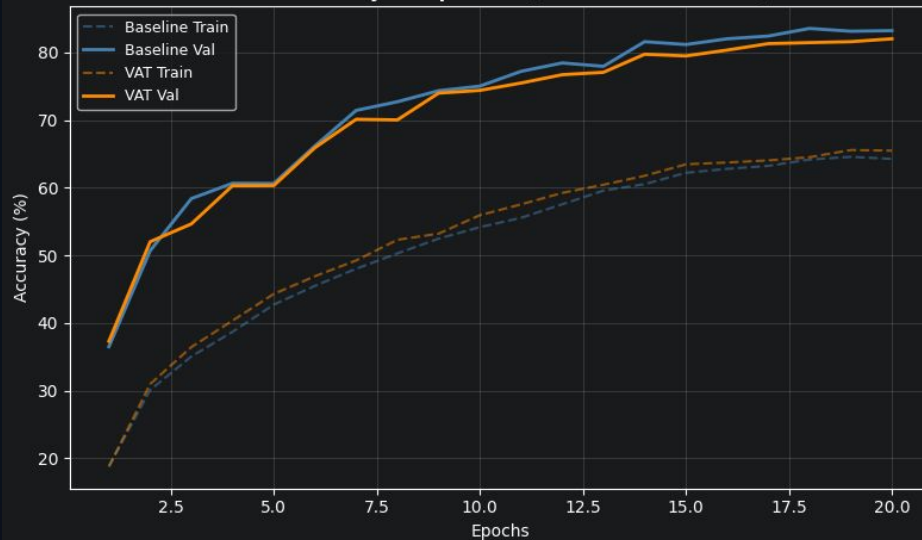
FCN Accuracy Comparison (0.0% Labeled Data)



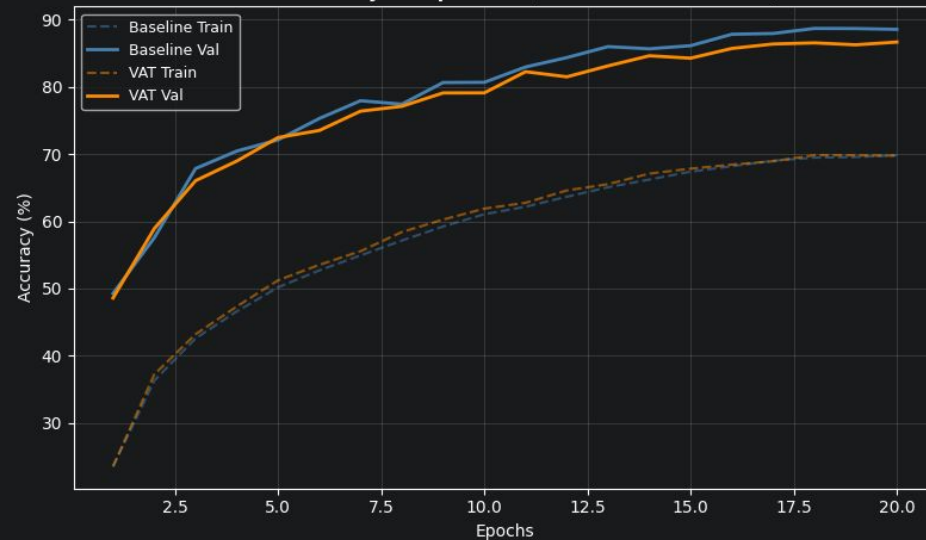
FCN Accuracy Comparison (25.0% Labeled Data)



FCN Accuracy Comparison (50.0% Labeled Data)



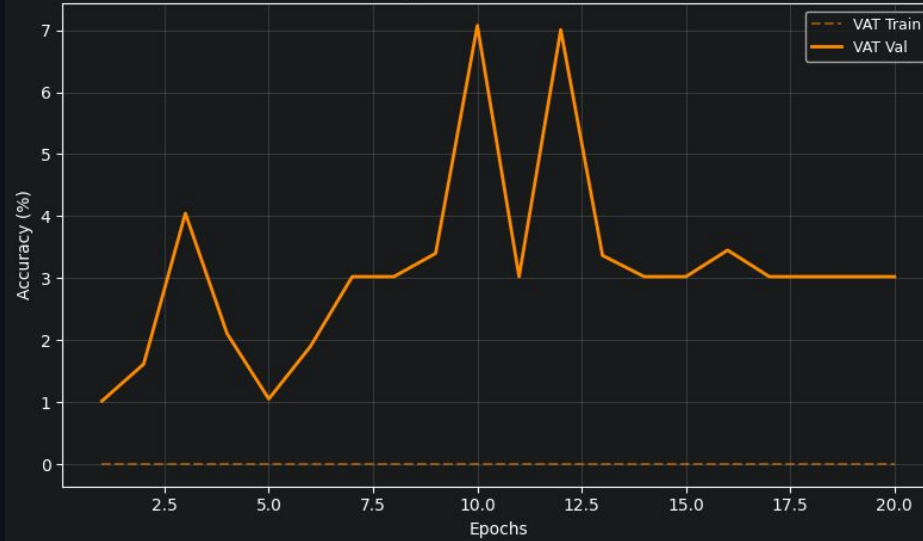
FCN Accuracy Comparison (100.0% Labeled Data)



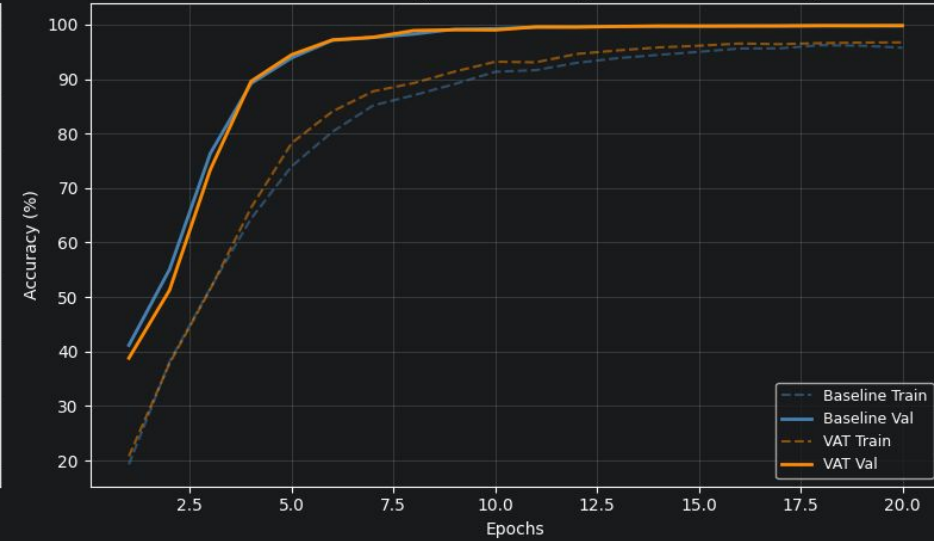
GTSRB: 0%, 25%, 50% and 100% Labeled Data

Small
CNN

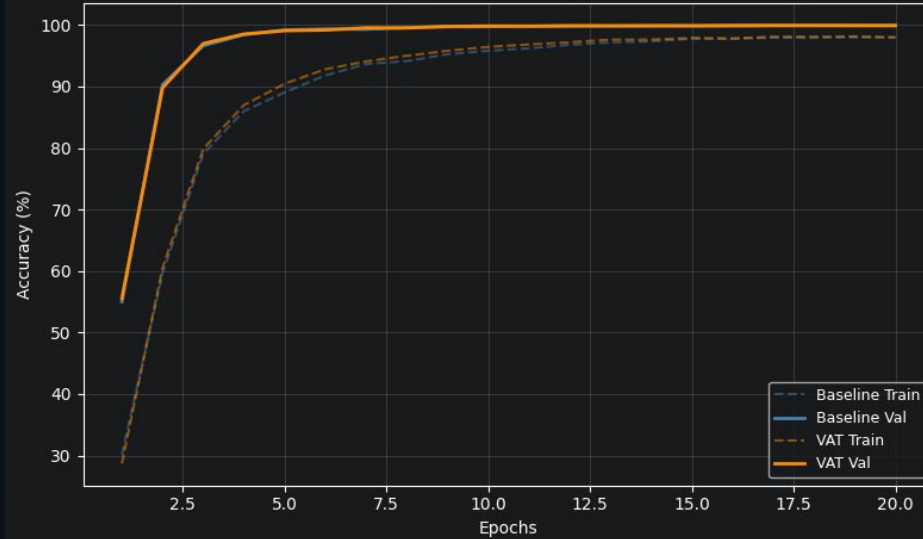
SmallCNN Accuracy Comparison (0.0% Labeled Data)



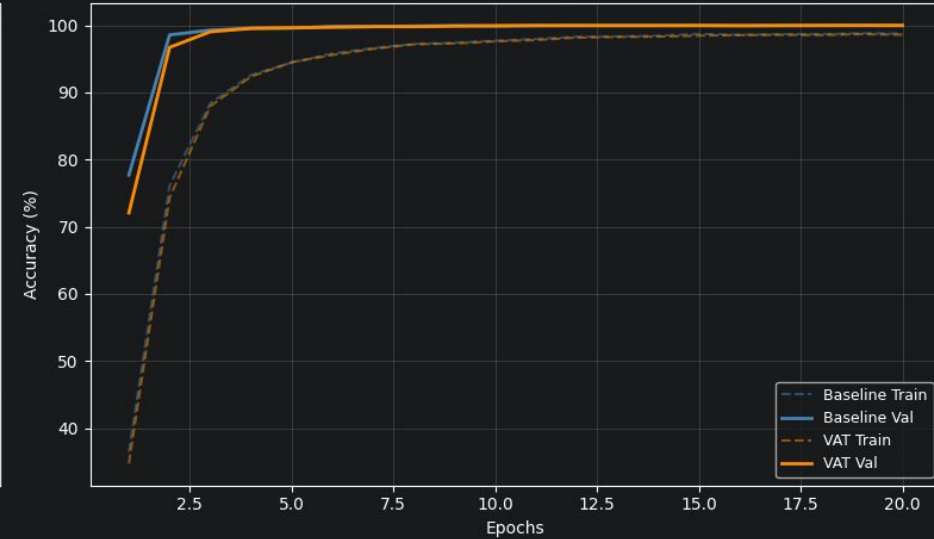
SmallCNN Accuracy Comparison (25.0% Labeled Data)



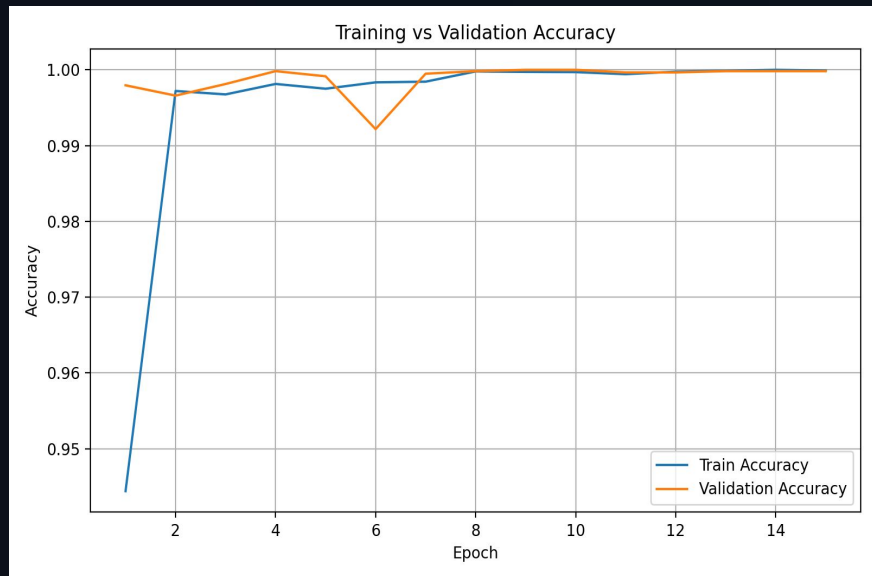
SmallCNN Accuracy Comparison (50.0% Labeled Data)



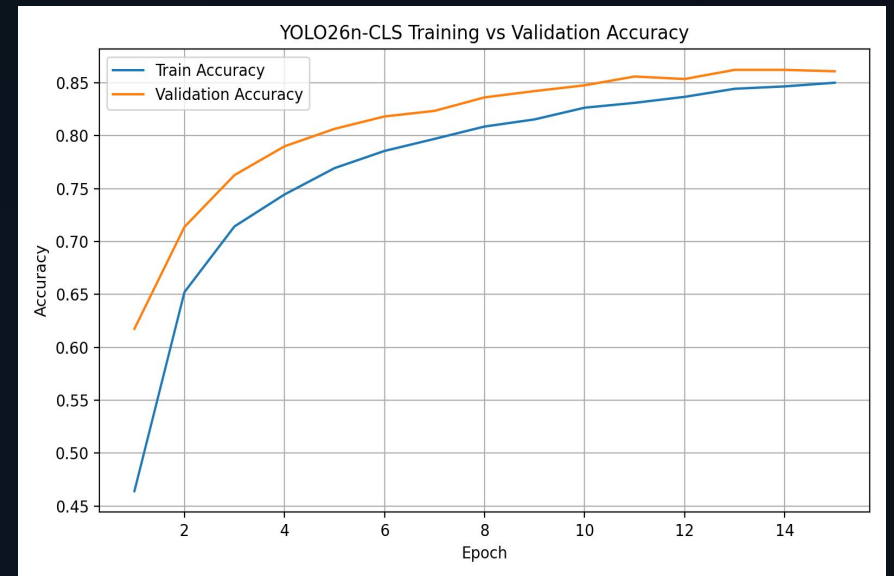
SmallCNN Accuracy Comparison (100.0% Labeled Data)



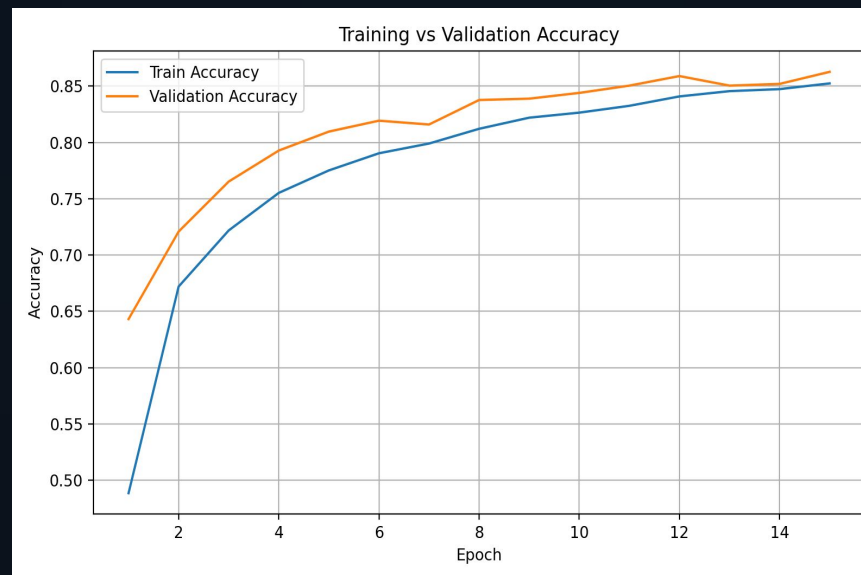
External Classification Benchmarks



ResNet18



YOLO11n



YOLO26n

The Conclusion

Experiment	Val Acc	Test Acc
FCN / baseline	96.45%	83.84%
FCN / vat	95.19%	82.20%
SmallCNN / baseline	100.00%	99.02%
SmallCNN / vat	100.00%	98.99%

~100%

SCNN Peak Accuracy

Convolution Dominates

The SCNN consistently outperformed the FCN across all experiments. This firmly validates the necessity of convolutional spatial feature extraction for complex image recognition tasks.

Virtual Adversarial Training (VAT) is highly effective when labels are scarce, accelerating convergence and significantly reducing the dependency on expensive manual data annotation.

Questions & Discussion

Thank you for exploring Semi-Supervised Learning in Autonomous Driving.

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